Circuit Analysis (Lab 7)

Experiment 7: Analysis of Operational Amplifiers Operations

Name: ……………………………. **Registration No:** …………………

Date: ……………………………. **Grade and Signature:** …………………

CLO4	Construct amplifiers with operational amplifiers.			
Psychomotor/ Affective	**Level1 (1)**	**Level2 (2-3)**	**Level3 (4-5)**	**Level4 (6-7)**
Report Marks (3)			**Total marks (10)**	

Objective

The objective of this lab is to investigate the input to output relationship for inverting and non-inverting amplifiers.

Apparatus Required

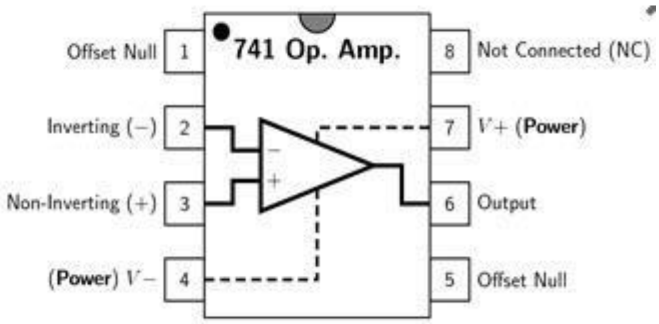
Sr. No.	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k Ω, 2k Ω ,3k Ω	3

Description

Operational amplifiers can be used in two basic configurations to create amplifier circuits. One is the inverting amplifier where the output is the inverse or 180° out of phase with the input, and the other is the non-inverting amplifier where the output is in the same sense or in phase with the input.

Both operational amplifier circuits are widely used and they find applications in different areas. When an operational amplifier or op-amp is used as a non-inverting amplifier it only requires a few additional components to create a working amplifier circuit.

Pin configuration of IC 741 op amp is shown below



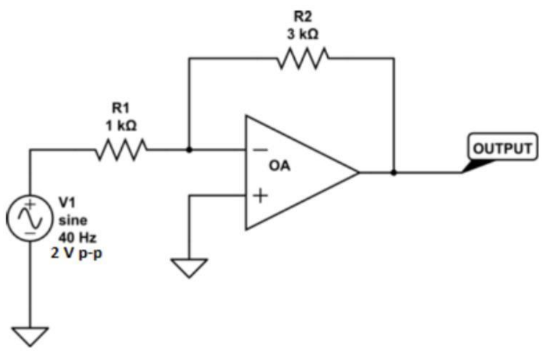
- Pin 1 is offset null.
- Pin 2 is inverting input terminal.
- Pin 3 is a non-inverting input terminal.
- Pin 4 is negative voltage supply (VCC)
- Pin 5 is offset null.
- Pin 6 is the output voltage.
- Pin 7 is positive voltage supply (+VCC)
- Pin 8 has no connection.

Precautions:

1. Voltage control knob should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position.
3. Carefully utilize the oscilloscope channels and function generator.

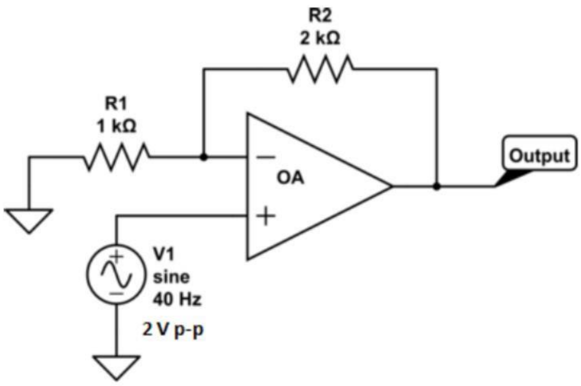
Procedure for Inverting Amplifier:

1. Wire up the circuit as shown in the figure below on the breadboard.
2. Give the input to the inverting terminal of the op-amp and simultaneously display the input and output on the oscilloscope.
3. Draw the input & output waveforms at the end of the lab.



Procedure for Non-Inverting Amplifier:

1. Wire up the circuit as shown in the figure below on the breadboard.
2. Give the input to the non-inverting terminal of the op-amp and simultaneously display the input and output on the oscilloscope.
3. Draw the input & output waveform at the end of the lab.



Result:

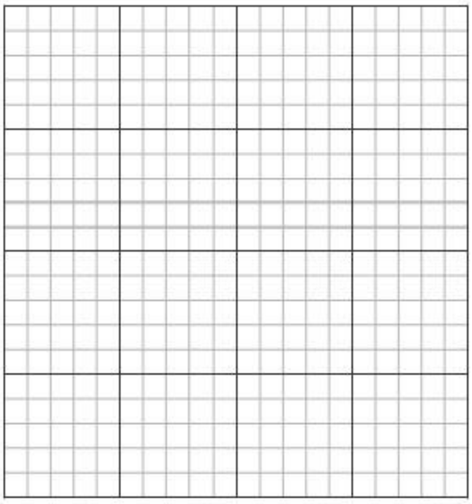


Figure 7: Input Voltage Signal

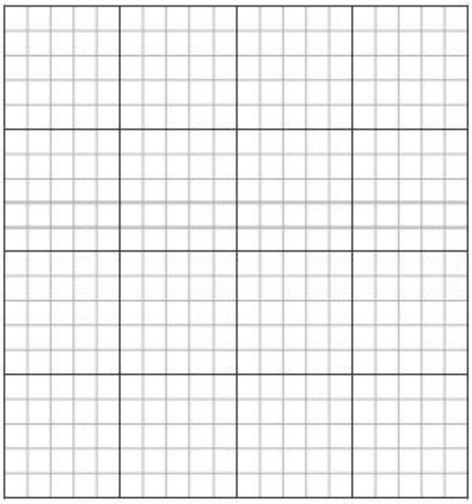


Figure 8: Inverted Output

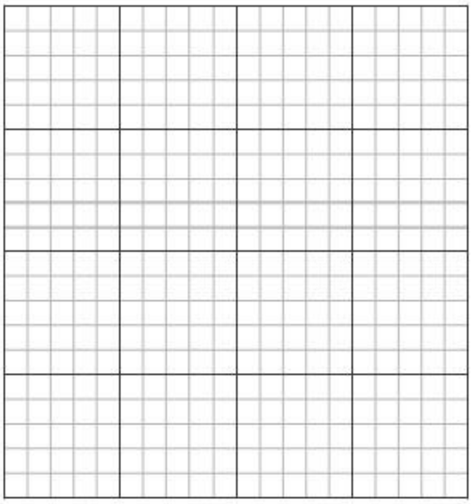


Figure 9: Non-Inverting Output

Post Lab Questions

1. Define inverting amplifier.

2. Define Non-inverting amplifier.

3. What are the applications of inverting amplifiers?

4. What are the applications of non-inverting amplifiers?

5. What is gain of an amplifier?

6. How do you calculate the gains for inverting and non-inverting amplifiers?